the scope of the magazine, we would like to introduce you to 'cron', a very powerful, yet simple task scheduler for Linux. Cron is distro-agnostic and comes bundled along with GNU/Linux, so you won't be required to separately install any packages. Cron is a system daemon, meaning that it runs unobtrusively in background and executes commands without user intervention. Commands or scripts that are required to be automated can be added to the 'crontab', a text file containing these details along with other details such as frequency/time interval between successive iterations of those commands. For e.g. you can have cron backup your personal files to an external disk or any networked location at a fixed interval. As an added feature, rsync can be used as a backup utility. Rsync is a program that performs differential copying. This means that rsync will only backup those files and folders that have changed since the last backup, speeding up the process considerably. So if you've got a repetitive task that you'd like to perform at a regular interval, cron provides great flexibility in specifying such an interval and automating the task for you.

There are two main ways of specifying the tasks to be automated using cron. First, users should find three folders in their /etc partition - cron.hourly, cron.daily and cron.monthly. As can be judged from the folder names, any script placed inside these folders (including symlinks) will automatically executed every hour, day or month depending on the folder. The second method is more powerful but slightly more complicated. Each user has a 'crontab' – a text file containing the commands to be executed along with information on when to execute them. To edit your crontab, open a terminal and type

```
crontab -e
```

The format of a crontab entry is as follows: minute (0-59) hour (0-23, 0 = midnight) day (1-31) month (1-12) weekday (0-6, 0 = Sunday) command. Each column entry is separated from the next with a single space. An asterisk `[*]` indicates that the task is to be executed at every instance of the time period. For eg, say we have a backup script located at `/home/user/Scripts/backup.sh`. If we want to run this script at 9 AM every Sunday, the crontab entry will look something like this:

```
* 9 * * 0 /home/user/Scripts/backup.sh
or
* 9 * * Sun /home/user/Scripts/backup.sh
```

In the second alternative, the '0' for day of the week has been replaced by 'Sun' which produces the same effect. As another example, if we want to automate something at 4:40pm everyday, our crontab entry will look something like this:

```
40 16 * * * /path/to/automation/script
```

Although the command prompt is a very powerful tool for Linux, some people are not very comfortable using it. For these people, there are plenty of GUI options available. Applications like Sikuli and Tasks Till Dawn are Java based apps (and are hence cross-platform) that can make the process of automation dead simple. Sikuli lets you automate virtually any task that involves interactions with a GUI application, by using simple screenshots of a portion of the interface and giving a corresponding action (Such as clicking or entering text). For more information on how to use Sikuli, visit http://dgit.in/O5w6WG

Android Automation put your mobile to work

Your phone can be configured to automate several mundane tasks which you perform on a daily basis. We show you how using Tasker and on{x}

So you've learned how to automate nearly any action on your computer or laptop, but the device that's most likely to be on you throughout the day is your smartphone. Smartphones have evolved at a dramatic pace over the years, and today's smartphones are just as powerful (if not better) than the laptops from a few years ago. Certainly all this power can be put to better use than throwing birds at pigs or making your high-res photo look like they were taken fifty years ago. In this workshop we show you how you can automate certain repetitive tasks on your phone and turn your handheld gadget into a true extension of your brain! Due to the open nature of the Android platform, Android phones have a near limitless potential for automation. If there's a task you'd like to automate, there's a very good chance that there's an app for that. During the course of this workshop, we take you through automating tasks on your phone with apps such as Tasker and other similar apps like Microsoft's new On{x} tool for building simple macros to automate your device. This includes things like backup your apps and settings automatically to your SD card, squeezing out a few extra hours from your battery using Juice Defender, and some futuristic stuff using NFC tags and your device's NFC chip. So let's get started!